Lab Day: Django CRUD + Auth Consolidation

Phase: Backend — Django
Prerequisites: Django modules `01` – `05` started (at least through auth forms)
Goal: Every learner finishes an **independent**, login-protected CRUD feature they can demo
Previous module: Security and Deployment
Next module: Lab Deploy Capstone Debug

Module Outcome:

- Finish one login-protected CRUD feature you can demo without the trainer driving
- Prove owner-only rows (Priya cannot delete Ravi's task)
- Leave with a GitHub commit that matches the Done when list

Lab brief (pick one track)

Track A — Task Manager: Task model (`title`, `is_done`, `owner` FK to User). List/create/toggle/delete only for logged-in owner.
Track B — Mini Blog: Post model (`title`, `body`, `author`). List/detail/create/edit; edit only if author.
Track C — CampusCart seed: Product list + "save favorite" for logged-in user (simpler than full cart).

Where used? Campus portal tasks/blog, CampusCart favorites, and any login-protected CRUD you will demo in placement.

Remember: Filter querysets by `owner=request.user` (or author) — otherwise every user sees everyone's data.

Must-pass checklist (exit criteria)

Done when

All boxes below must be checked before demo day.

- [ ] `runserver` works; migrations applied
- [ ] Register → login → logout works
- [ ] Anonymous user redirected from create/edit
- [ ] CRUD works for own records; no crash on empty list
- [ ] CSRF on all POST forms
- [ ] Flash messages on success
- [ ] Admin can view models
- [ ] GitHub push with standard deliverables (README, requirements, `.env.example`, screenshots)

Common stuck points

Symptom	Hint
404 on `/tasks/`	App urls included in project urls? trailing slash?
Form saves but blank owner	Set `form.instance.owner = request.user` before save
Everyone sees all tasks	Filter queryset by `owner=request.user`
CSRF 403	`{% csrf_token %}` inside `<form method="post">`
Redirect loop	LOGIN_URL / login_required mismatch

Skill-level rubric (today)

Level	Evidence
Not started	No runnable app
Guided	Works only with help
Independent	Completes must-pass checklist alone
Interview-ready	Explains owner filtering, CSRF, and session in plain English

Assignment

Finish any incomplete checklist items + push repo. Demo video 2–3 min.

Expected: README shows login screenshot + task list screenshot; clone instructions work for a peer.